Q.2.

2. Check that $3\,du^2 + 6\,du\,dv + 5\,dv^2$ defines an inner product on $\mathbb{R}^2$. Find the angle between $\begin{pmatrix}1\\0\end{pmatrix}$ and $\begin{pmatrix}1\\1\end{pmatrix}$ with respect to this inner product.

Prop. 45 $E\,du^2 + 2F\,du\,dv + G\,dv^2$

$E = 3,\ F = 3\ \ G = 5$

$\begin{pmatrix} E & F \\ F & G \end{pmatrix} = \begin{pmatrix} 3 & 3 \\ 3 & 5 \end{pmatrix}$ which is positive definite.

(because $E = 3 > 0$

$EG - F^2 = 15 - 9 = 6 > 0$)

$$g(\vec{v},\vec{w}) = \vec{v}^T \begin{pmatrix} E & F \\ F & G \end{pmatrix} \vec{w}$$

write $g = E\,du^2 + 2F\,du\,dv + G\,dv^2$.

} Same Thing!

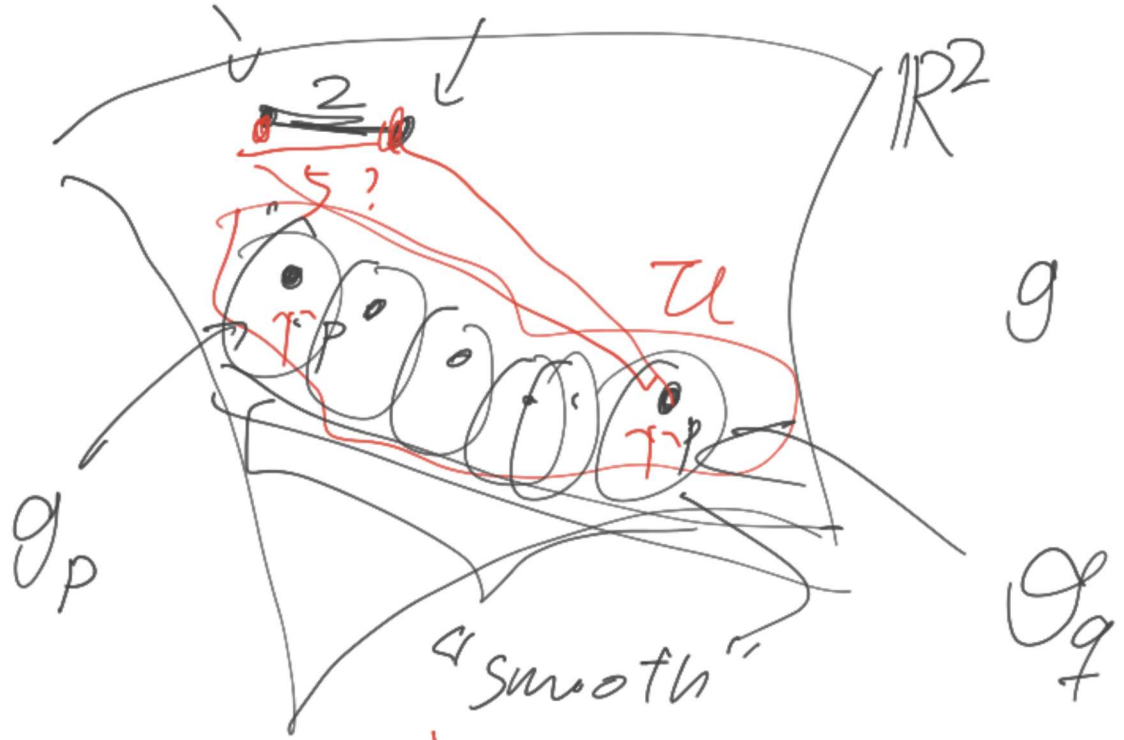
Prop. 4.5 g defines an inner product on $\mathbb{R}^2$ or U iff

$\begin{pmatrix} E & F \\ F & G \end{pmatrix}$ is positive definite.

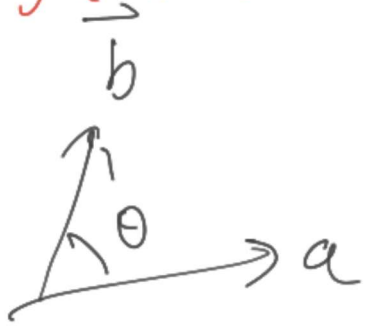
for all $p \in U$.

$$g\left(\begin{pmatrix}1\\0\end{pmatrix},\begin{pmatrix}1\\1\end{pmatrix}\right) = (1\ 0)\begin{pmatrix}3 & 3\\ 3 & 5\end{pmatrix}\begin{pmatrix}1\\1\end{pmatrix} = (1\ 0)\begin{pmatrix}6\\8\end{pmatrix} = 6,$$

$\cos\theta = \quad g($



$$\|a\| = \sqrt{g(a,a)}$$



$$\cos\theta = \frac{\vec{a}\cdot\vec{b}}{\|\vec{a}\|\,\|\vec{b}\|}$$

$$= \frac{g(\vec{a},\vec{b})}{\sqrt{g(\vec{a},\vec{a})\,g(\vec{b},\vec{b})}}$$

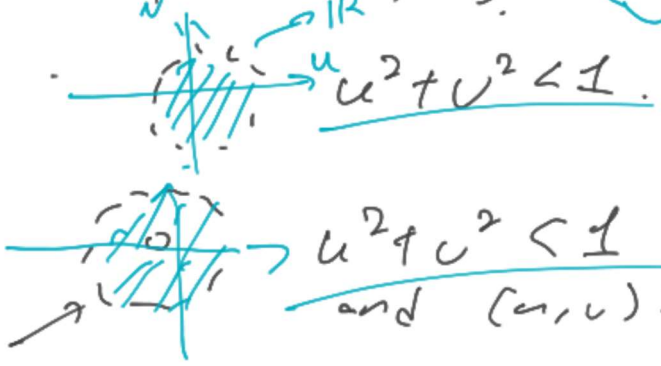
$(1\ 0)\begin{pmatrix}3&3\\3&5\end{pmatrix}\begin{pmatrix}1\\0\end{pmatrix}$ choice.

$$\frac{g\left(\begin{pmatrix}1\\0\end{pmatrix},\begin{pmatrix}1\\1\end{pmatrix}\right)}{\sqrt{g\left(\begin{pmatrix}1\\0\end{pmatrix},\begin{pmatrix}1\\0\end{pmatrix}\right)}\,\sqrt{g\left(\begin{pmatrix}1\\1\end{pmatrix},\begin{pmatrix}1\\1\end{pmatrix}\right)}}$$

Defn Riemannian Metric g on $U \subseteq \mathbb{R}^2$

$\{(u,v) \in \mathbb{R}^2 \mid (\dots)\}$





$x^2+y^2=1$

① Formal Way:

For every $\vec{p} \in U$, we have an inner product: $g_p : \mathbb{R}^2 \times \mathbb{R}^2 \to \mathbb{R}$

("Smoothness requirement")

② "Working" Way

Write

$g = E\,du^2 + 2F\,du\,dv + G\,dv^2$

where E, F, G depends on $\vec{p}$.

(E, F, G are functions $U \subseteq \mathbb{R}^2 \to \mathbb{R}$ & $\begin{pmatrix} E & F \\ F & G \end{pmatrix}$ pos def.)

$$g_p(\vec{v},\vec{w}) = \vec{v}^T \begin{pmatrix} E_p & F_p \\ F_p & G_p \end{pmatrix} \vec{w}$$

Real Numbers!